

LegalEase AI Platform

Complete System Blueprint
and Feature Diagrams | v3.0

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Indian Legal AI | Next.js + FastAPI + Ollama + Gemini

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1. Executive Summary

Details

LegalEase is an Indian legal AI assistant for law firms and solo practitioners. It combines local document RAG (your PDFs), live web legal search (Gemini), deep hybrid research reports, and a continuous self-improvement loop.

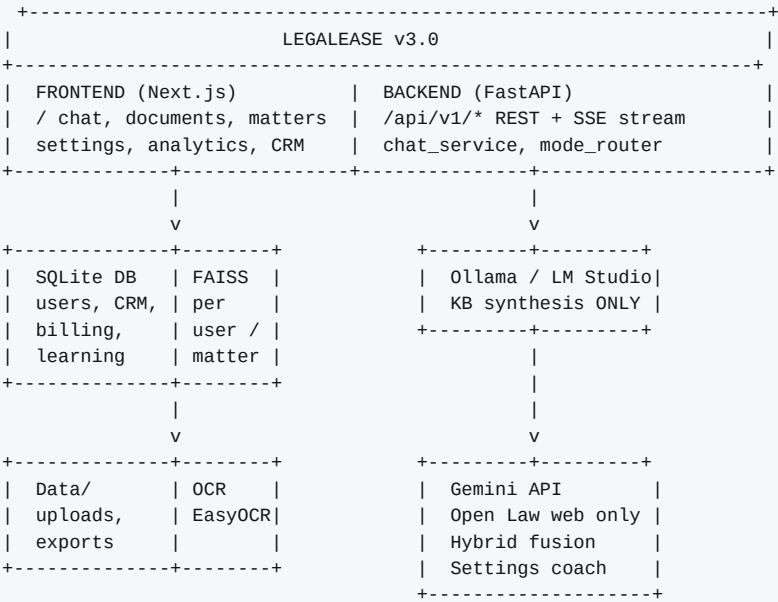
Core design rule: Knowledge Base answers come ONLY from uploaded documents + local Ollama/LM Studio. Gemini never writes KB answers.

Stack: Next.js frontend | FastAPI backend | FAISS vector store | Ollama/LM Studio local LLM | Gemini API (Open Law + Hybrid fusion + Settings coach only).

Deployment model: Single-lawyer self-hosted (production-ready). Multi-tenant cloud requires additional per-user model routing (not required for solo use).

2. Master System Architecture

Diagram



Details

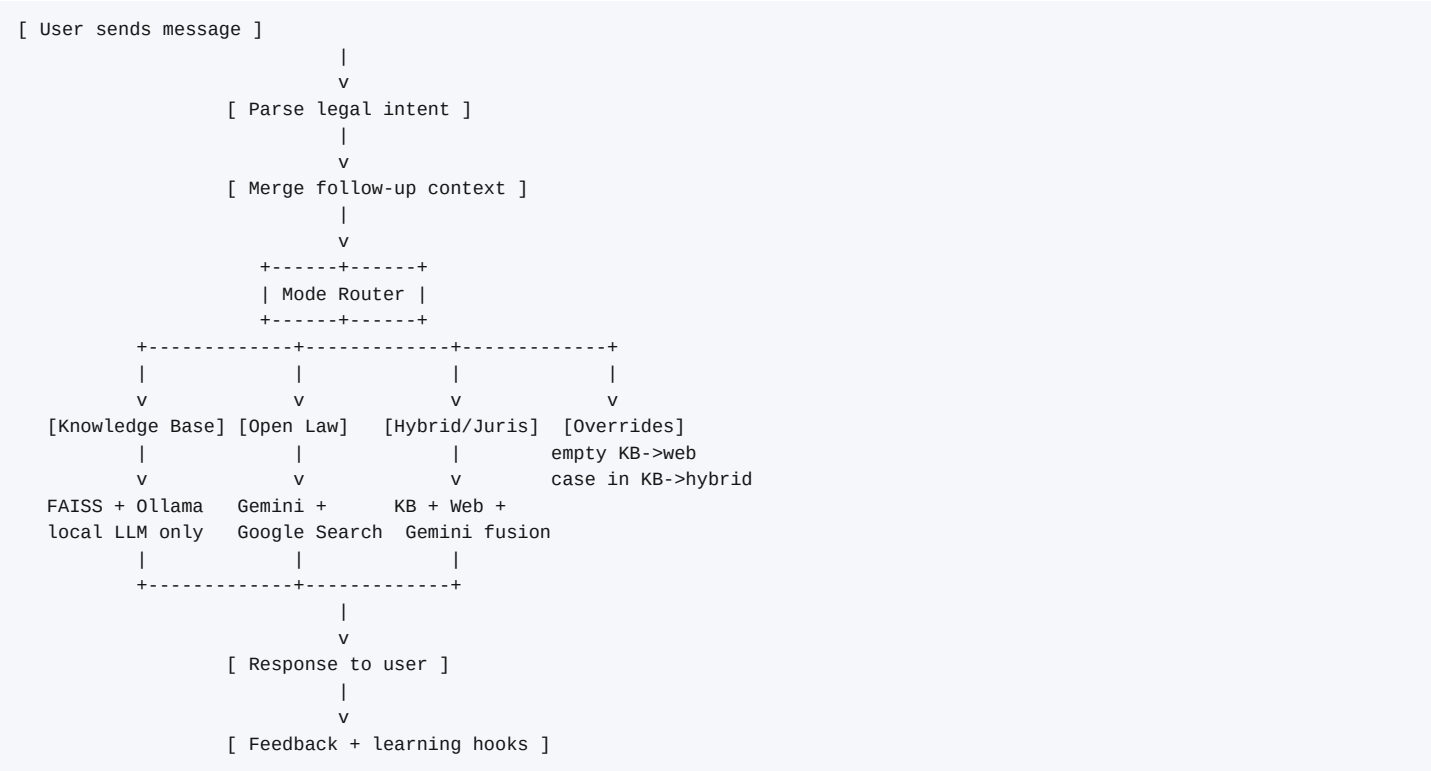
Data flow: User -> Next.js -> FastAPI -> Mode Router -> (KB | Open Law | Hybrid)
-> Local LLM and/or Gemini -> Response -> Learning hooks -> Background automation.

Key entry files:

- backend/app/main.py API startup, middleware, schedulers
- backend/app/services/chat_service.py Chat orchestration
- backend/app/services/mode_router.py Mode selection
- rag.py / kb_pipeline.py RAG retrieval pipeline
- llms.py Ollama / LM Studio client
- web/app/(app)/page.tsx Main chat UI

3. Three Chat Modes - Routing Diagram

Diagram



Details

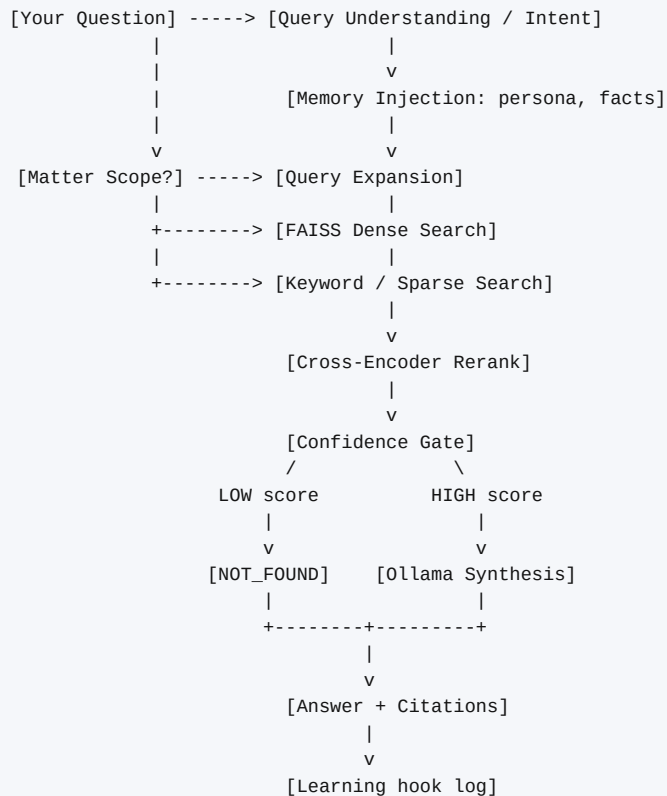
Mode	UI Name	Engine	Best for
knowledge_base	Knowledge Base	FAISS + Ollama	Your uploaded documents
web_search	Open Law	Gemini + web search	Live statutes, cases, news
deep_case	Hybrid (Pro+)	KB + Gemini fusion	Full research reports

Smart overrides (mode_router.py):

- Empty KB + case question -> may suggest Open Law
- KB mode + case explanation -> may upgrade to Hybrid (Pro membership)

4. Knowledge Base Engine - Feature Diagram

Diagram



Details

Steps:

1. Query understanding - section lookup, comparison, list, case intent
2. Memory injection - persona, facts, coach style (never legal substance from Gemini)
3. Hybrid retrieval - dense FAISS + sparse keyword + optional cross-encoder rerank
4. Confidence gate - low scores return NOT_FOUND instead of hallucination
5. Synthesis - local LLM writes answer from chunks only (GEMINI_KB_SYNTHESIS=0)
6. Per-matter scoping - matter_id limits search to that case file's documents

Indexing: Upload PDF/image -> OCR if scanned -> chunk -> embed -> FAISS index

Path: faiss_indexes/user_{id}/matter_{matter_id}/

Key files: kb_pipeline.py, rag.py, answer_orchestrator.py, llms.py, matter_index.py

5. Open Law (Web Intelligence) - Feature Diagram

Diagram

```

[User: Open Law question]
  |
  v
[Filter KB/Hybrid history] <-- prevents document leak
  |
  v
[Classify depth: quick | standard | detailed | comparison]
  |
  v
[Build self-contained query]
  |
  v
[Gemini + Google Search Grounding]
  |
  v
[KB-leak detection] ---- retry without history if doc dump detected
  |
  v
[Format: bullets | table | short fact]
  |
  v
[Response + source URLs + trust badges]

```

Details

Adaptive response sizing (web_intelligence.py):

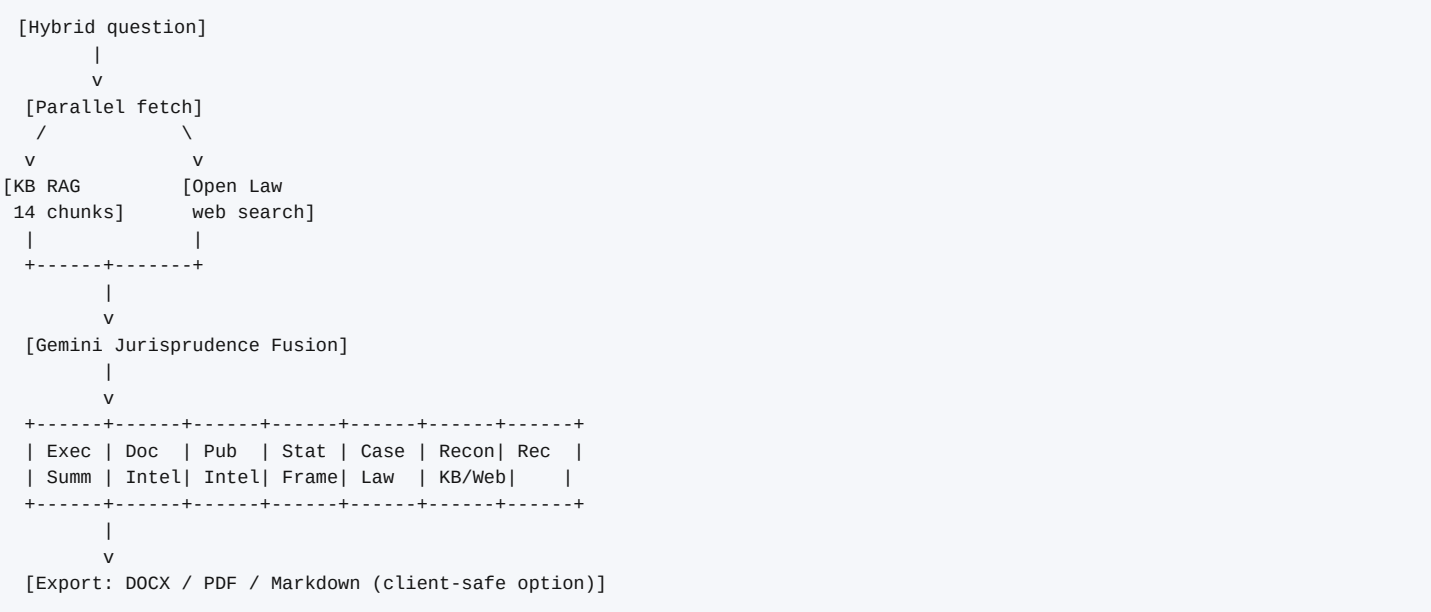
- Quick fact ("Who is CJI?") -> 2-4 sentences
- Standard question -> concise bullets (~260 words)
- "Explain in detail" -> structured sections (~480 words)
- Comparison ("IPC 300 vs 307") -> markdown comparison table

Safety: No KB history sent to Gemini. No "Gemini" branding in UI.

Gemini daily quota enforced by plan (Free 15 / Pro 200 / Legal Pro 1000).

6. Hybrid / Jurisprudence Engine - Feature Diagram

Diagram

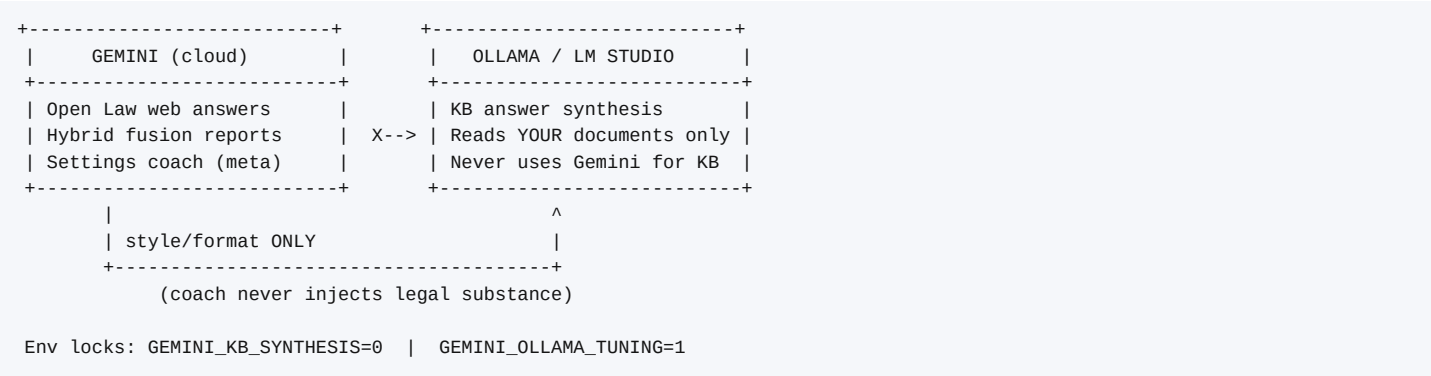


Details

Conflict rule: Uploaded documents win when KB and web disagree on sections/statutes.
Requires Pro or Legal Pro membership (Free users downgraded to KB).
Includes contradiction checker and citation verifier on output.

7. Gemini vs Ollama - Separation of Duties

Diagram



Details

Task	Engine
Answer from uploaded PDFs	Ollama only
Live web legal search	Gemini only
Hybrid research report	Both
Tune Ollama style/retrieval	Gemini coach
Answer legal Q in coach	FORBIDDEN

8. AI Memory & Persona

Diagram

```
[Settings: AI Memory]
|
+-- Persona (warm / professional / concise / detailed)
+-- Practice area (e.g. Criminal litigation)
+-- Facts (key-value, manual + auto-suggested)
+-- Communication notes
+-- Coach memories (style only)
+-- Thread summaries
|
v
[Injected every chat turn]
|
v
[KB + Open Law + Hybrid prompts]
```

Details

Auto-facts appear in amber badges - user can edit/delete wrong ones.

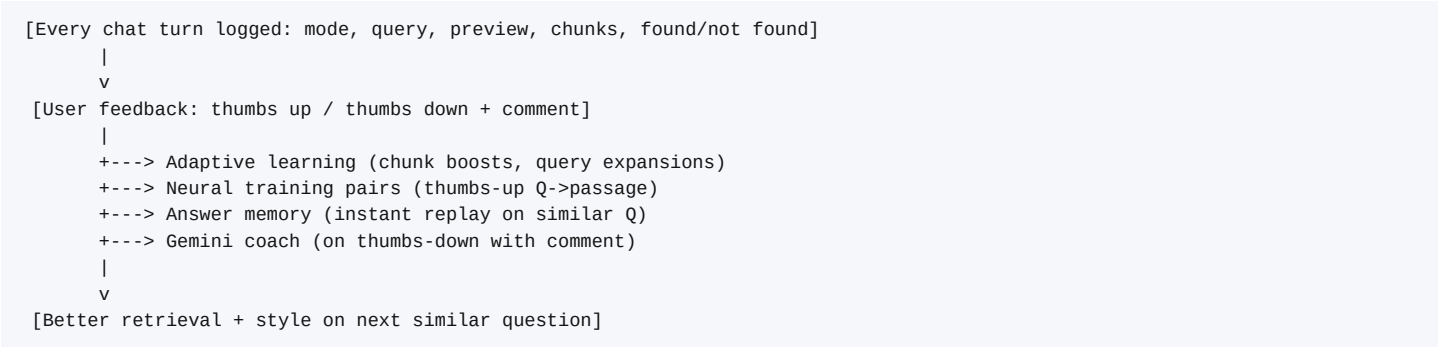
Past chat RAG: semantic search over prior threads ("what did we conclude about...").

API: /api/v1/memory - profile, facts CRUD, reindex chats.

Key file: backend/app/core/user_memory.py

9. Learning & Feedback Loop

Diagram



Details

Thumbs up: boosts chunks, adds neural pair, answer memory, triggers automation pipeline.

Thumbs down + comment: records complaint, triggers coach analysis if enabled.

Mode stats track hit rate per KB / Open Law / Hybrid.

Key files: adaptive_learning.py, learning_engine.py

Diagram



Details

Coach outputs ALLOWED: persona suggestion, style facts, search phrasing, verified training pairs.

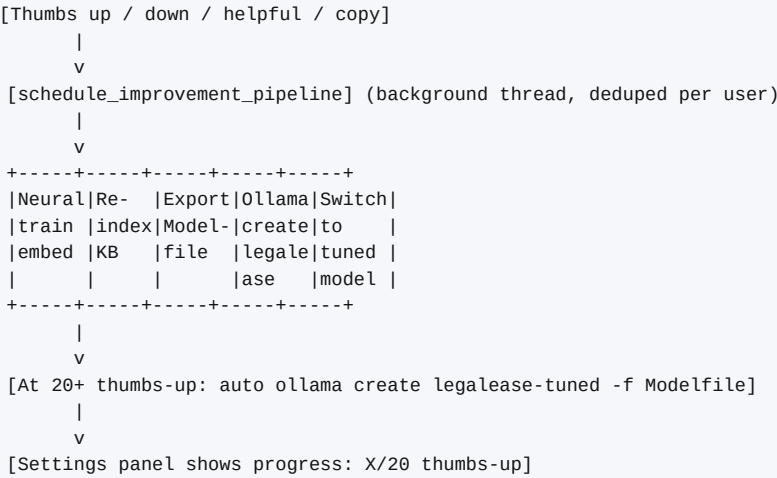
Coach outputs FORBIDDEN: legal holdings, IPC/BNS text, Gemini-generated training pairs.

Schedule defaults: 1 day interval, 1 min new feedback, check every 30 min.

Key files: gemini_ollama_coach.py, coach_scheduler.py

11. Auto-Improvement Pipeline (Fully Automated)

Diagram



Details

Env vars: IMPROVEMENT_AUTO=1, OLLAMA_AUTO_REINDEX=1, OLLAMA_AUTO_CREATE=1, OLLAMA_AUTO_USE_TUNED=1, OLLAMA_AUTO_EXPORT_MIN_THUMBS=20

Export path: Data/ollama_exports/{user_id}/{timestamp}/

Files: Modelfile, training.jsonl, README.txt

Log: Data/ollama_exports/automation_log.jsonl

Key file: backend/app/core/improvement_automation.py

12. Neural Training & Modelfile Export

Diagram

```
LEVEL 1 - Embedding fine-tune:  
  [Thumbs-up pairs] -> [SentenceTransformer train] -> [Re-index FAISS]  
  
LEVEL 2 - Modelfile export:  
  [20+ thumbs-up] -> [Export Modelfile + JSONL] -> [ollama create legalease-tuned]  
  
Both compound over time for better retrieval AND better answer style.
```

Details

Neural: NEURAL_FINETUNE_AUTO=1, min 4 pairs, trains MiniLM variant on Q->passage pairs.

Modelfile: base model + system prompt + few-shot examples from verified feedback.

After embedding train: MUST re-index documents for new embeddings to take effect.

13. Documents & Knowledge Base Health

Diagram

```
[/documents page]
|
+-- Drag-drop upload (PDF, images)
+-- Assign to matter
+-- OCR for scans (OCR_ENABLED=1)
+-- Entity extraction + timeline
+-- Delete / reindex
|
v
[KbHealthPanel: embeddings online, FAISS count, stale warnings]
|
v
[Auto-reindex if index empty but docs exist]
```

Details

API: /api/v1/documents - upload, list, index, kb/reindex-auto, entities, timeline.

Per-matter upload: ?matter_id= scopes index to that case file.

14. Matters & Case Scoping

Diagram

```
[Create Matter / Case File]
|
+-- Link documents
+-- Notes
+-- Client portal token link
+-- E-sign mock flow
|
v
[Chat: select Case File before KB/Hybrid question]
|
v
[FAISS searches ONLY that matter's index]
```

Details

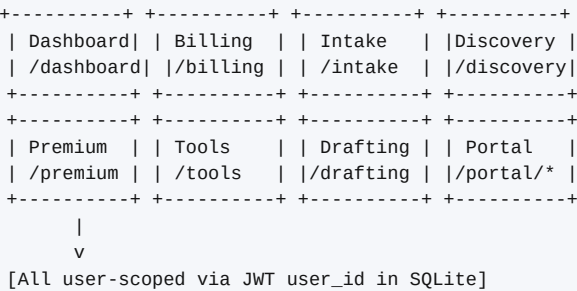
Index path: faiss_indexes/user_{id}/matter_{matter_id}/

Unlinked docs: faiss_indexes/user_{id}/_unlinked/

Matter autopilot: entity extraction + suggested research queries.

API: /api/v1/matters

Diagram



Details

Module	Route	Purpose
Dashboard	/dashboard	Practice overview, stats
Billing	/billing	Time entries, GST invoices
Intake	/intake	CRM lead capture
Discovery	/discovery	E-discovery triage + job queue
Premium	/premium	Witness sim, BNS audit, deal rooms
Tools	/tools	IPC lookup, fees, contract tools
Drafting	/drafting	Template drafting studio
Portal	/portal/*	Client-facing matter view
E-sign	/esign/mock	Mock signing flow

16. Premium Tools Detail

Diagram

```
[Premium /premium]
|
+-- Witness / Mock Trial Simulator (disposition: evasive, hostile...)
+-- Precedent Tree (citation graph + judge analytics)
+-- BNS Auditor (IPC->BNS risk scan on text/docs)
+-- Deal Rooms (multi-doc diligence + anomaly detection)
+-- PII Redaction (detect, whitelist, redact by entity type)
+-- Redline Engine (clause revision with diff HTML)
```

Details

API: /api/v1/premium - witness sessions, BNS audit, deal rooms, PII, redline.
Separate premium learning store for tool feedback.

17. Billing, CRM & Trust

Diagram

```
BILLING (/billing):  
  [Time entry] -> [AI narrative polish] -> [Invoice + GST] -> [Summary]  
  
CRM (/intake):  
  [Lead] -> [AI intent classify] -> [Pipeline stages] -> [Follow-up email draft]  
  [Public intake form] -> unauthenticated lead capture  
  
TRUST (/billing ledger):  
  [Matter trust balance] <-> [Transaction ledger]
```

Details

Billing API: /api/v1/billing - time entries, invoices, narrative polish, lexicon.

CRM API: /api/v1/crm - leads, classification, corrections, follow-ups.

Trust API: /api/v1/trust - matter trust balances and transactions.

Subscription billing: mock upgrade (Stripe not wired); practice billing is fully implemented.

18. E-Discovery & Research

Diagram

```
[Upload batch to matter]
  |
  v
[Background job queue (Redis/SQLite)]
  |
  v
[Triage: relevance / privilege-style scoring]
  |
  v
[Search within batch + tag/classify items]
  |
  v
[Job status polling in /discovery UI]
```

Details

API: /api/v1/ediscovery - batches, jobs, triage, search, review.

Worker: scripts/ediscovery_worker.py (optional background process).

Research log: /api/v1/research - query expansion, history, feedback.

19. Settings, Engine Status & Analytics

Diagram

SETTINGS (/settings):
AI Memory | Coach | Auto-improvement | Profile | LLM test | OCR status

ENGINE STATUS BAR (chat top):
[KB chips] [Web Intel] [LLM online] [Memory active] [Gemini quota]
Polls every ~45 seconds

ANALYTICS (/analytics):
Mode stats | Learned queries | Deal rooms | Witness sessions | JSONL export

Details

Settings sections: persona, facts, neural train, Ollama coach, auto-improvement pipeline, daily auto-coaching toggle, Modelfile export, LLM backend status.

Analytics API: /api/v1/learning/analytics/full

20. Auth, Membership & Rate Limits

Diagram



Details

Auth: backend/app/core/auth.py, legacy_saas/legalease_auth.py

Rate limits: backend/app/middleware/rate_limit.py (in-memory, per IP/token prefix)

Membership stored in JWT + users table.

21. Frontend Pages Map

Details

/login	Authentication
/	Main AI chat (KB / Open Law / Hybrid)
/documents	Upload & index knowledge base
/matters	Case files
/settings	Memory, coach, LLM config, auto-improvement
/analytics	Learning & usage stats
/billing	Time & billing + trust ledger
/premium	Advanced tools
/tools	Legal calculators & lookups
/discovery	E-discovery
/intake	CRM
/drafting	Document drafting studio
/dashboard	Practice dashboard
/portal/[token]	Client portal (read-only matter view)
/esign/mock/*	Mock e-signature flow

22. API Reference (All Groups)

Details

Base URL: /api/v1

Prefix	Purpose
/chat	Send message, stream SSE, export report
/learning	Feedback, stats, neural, coach, automation
/memory	Profile, facts, reindex chats
/documents	Upload, list, KB health, reindex
/engines	Status bar, watchlist, autopilot
/matters	Case CRUD, notes, docs
/sessions	Thread history, attachments
/premium	Witness, BNS audit, deal rooms, PII
/billing	Time entries, invoices, narratives
/crm	Leads, intake classification
/trust	Trust ledger
/ediscovery	Review batches, jobs
/research	Research log
/templates, /clauses	Drafting templates and clause library
/portal	Client portal
/esign	E-signature requests
/practice	Dashboard overview, public intake
/health	Live probe (no ML import)

23. Environment Variables (Essential)

Details

Local KB brain

LLM_BACKEND=ollama

OLLAMA_MODEL=gemma3:4b

OLLAMA_BASE_URL=http://127.0.0.1:11434

Web intel + coach

GEMINI_API_KEY=your_key

GEMINI_OLLAMA_TUNING=1

GEMINI_KB_SYNTHESIS=0 # MUST stay 0

Learning & automation

LEARNING_ENGINE_ENABLED=1

NEURAL_FINETUNE_ENABLED=1

NEURAL_FINETUNE_AUTO=1

IMPROVEMENT_AUTO=1

OLLAMA_AUTO_REINDEX=1

OLLAMA_AUTO_CREATE=1

OLLAMA_AUTO_USE_TUNED=1

OLLAMA_AUTO_EXPORT_MIN_THUMBS=20

OLLAMA_TUNED_MODEL_NAME=legalease-tuned

Daily coach

COACH_AUTO_SCHEDULE=1

COACH_AUTO_INTERVAL_DAYS=1

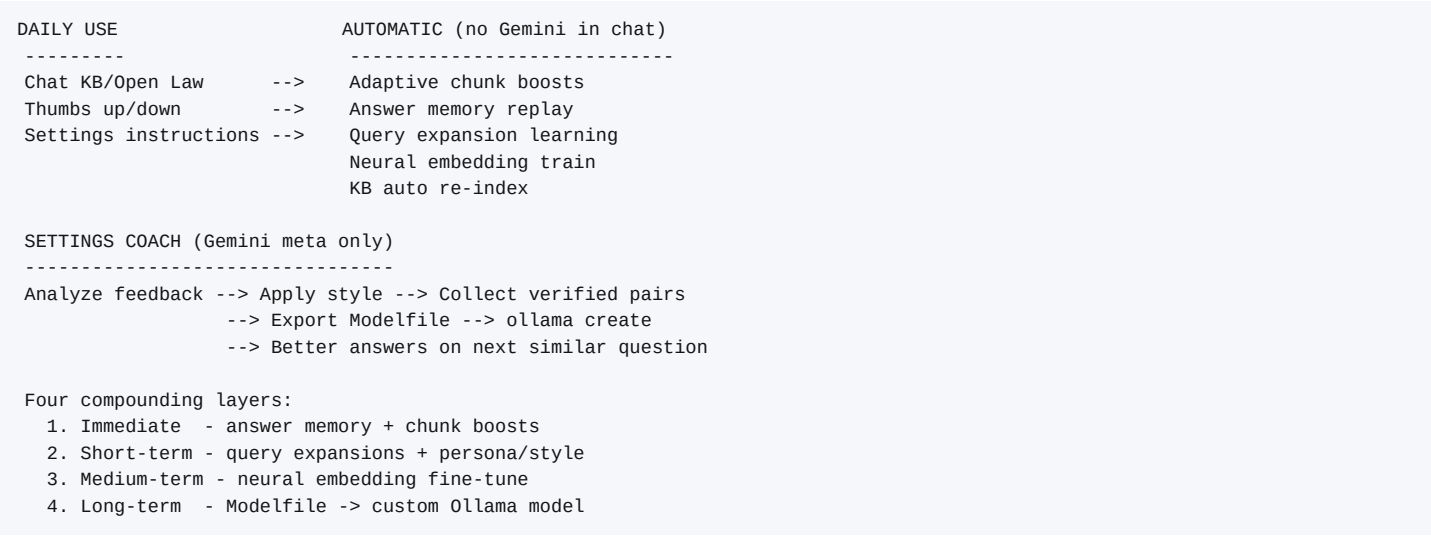
COACH_AUTO_MIN_NEW_FEEDBACK=1

COACH_AUTO_CHECK_INTERVAL_SEC=1800

See .env.example for full RAG, OCR, rate limit, and Redis settings.

24. Continuous Improvement Loop (Master)

Diagram



Details

Gemini's role is coach only - never answers KB questions, cannot inject legal substance.
Automation runs in background threads; never blocks chat.

25. Project File Structure

Details

Legal_AI_Final 3/	
backend/app/	FastAPI application
api/v1/endpoints/	REST route handlers (19 modules)
core/	Business logic (50+ modules)
services/	Chat, hybrid, mode routing
web/	Next.js frontend
app/(app)/	Main pages (chat, documents, settings...)
components/	React UI components
lib/api.ts	API client
rag.py	Core RAG retrieval
kb_pipeline.py	Multi-stage KB pipeline
llms.py	Ollama / LM Studio client
app.py	Legacy Streamlit monolith
legacy_saas/	Auth, legacy API routes
Data/	Uploads, FAISS indexes, exports
docs/	Documentation + this blueprint export
scripts/	Workers, eval, blueprint generator
tests/	Pytest suites (phases 1-4, SaaS, automation)
deploy/	Docker, nginx TLS config

26. Quick Start Checklist

Details

```
[ ] Ollama or LM Studio running locally
[ ] GEMINI_API_KEY in root .env
[ ] GEMINI_OLLAMA_TUNING=1 and GEMINI_KB_SYNTHESIS=0
[ ] LLM_BACKEND=ollama (for tuned model support)
[ ] ollama CLI on PATH (for auto-create)
[ ] Upload documents on /documents
[ ] Enable coach in Settings -> AI memory
[ ] Chat and use thumbs-up/down with comments
[ ] At 20+ thumbs-up: auto Modelfile export + ollama create legalease-tuned
[ ] Re-index after neural embedding training (automatic when IMPROVEMENT_AUTO=1)
[ ] Optional: run ediscovery worker for background discovery jobs
```